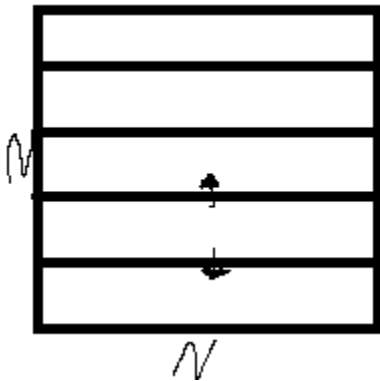


Q2

Assumptions

- 2D data set with N_{grid} total grid points, N on each side.
- χ FLOPs per iteration for each grid point.
- P processors, that can perform S FLOPs a second.
- A latency L and bandwidth B .
- Σ grid points on boundaries between partitions.
- Only one comm operation per iteration
- No overlap between comms and computation
- Size of each message is σ bytes per grid point.
- Periodic boundary conditions

1D Data Decomposition



For 1D decomposition, the grid is split into strips of size $\frac{N^2}{P}$. There are only 2 communications per iteration per strip, and each boundary has N points.

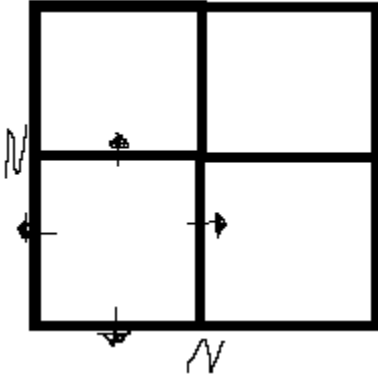
Hence, $\Sigma = 2N$.

Figure of Merit, $F = \frac{P\Sigma}{N_{grid}} = \frac{P\Sigma}{N^2} = \frac{2P}{N}$ i.e. the fraction of grid points involved in comms is of order P .

The max number of processors before speed does not increase is given by $P_{max} = \frac{\chi N_{grid} B}{(\sigma \Sigma + N_m B L) S}$ for N_m the number of messages.

For 1D, $P_{max} = \frac{\chi N^2 B}{(2\sigma N + N_m BL)S}$

2D Data Decomposition



For 2D decomposition, the grid is split into subgrids of size $\frac{N^2}{P}$. There are now 4 communications per iteration per subgrid, and each boundary has $\frac{N}{\sqrt{P}}$ points.

Hence, $\Sigma = 4 \frac{N}{\sqrt{P}}$.

Figure of Merit, $F = \frac{P\Sigma}{N_{grid}} = \frac{P\Sigma}{N^2} = \frac{4\sqrt{P}}{N}$ i.e. the fraction of grid points involved in comms is of order $\sqrt{P}$.

$$P_{max} = \frac{\chi N_{grid} B}{(\sigma \Sigma + N_m BL)S}$$

For 2D, $P_{max} = \frac{\chi N^2 B}{(4\sigma N \sqrt{P} + N_m BL)S}$

Efficiency

2D decomposition will become more efficient when its figure of merit is lower than the 1D decomposition.

Hence:

$$\frac{4\sqrt{P}}{N} < \frac{2P}{N}$$

$$P > 4$$

Hence when there are more than 4 processors, the 2D decomposition is more efficient.